Mirza Julkawsar

Software Engineer | AI / NLP / LLM Engineer

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PROFESSIONAL SUMMARY

CSE graduate from Brac University with a strong academic record and leadership experience, specializing in NLP, LLM engineering and full-stack development. Passionate, quick learner who has led and built end-to-end LLM and RAG systems (fine-tuning, hybrid retrieval, multi-agent orchestration) and authored a thesis on pragmatic inference in language models. Seeking a software, AI/ML or NLP engineer role where I can turn ideas into working products.

TECHNICAL SKILLS

Programming Languages: Python, Java, C, JavaScript, R, PHP, x86 Assembly (8086)
AI / ML / NLP: PyTorch, TensorFlow, scikit-learn, pandas, NumPy, HuggingFace Transformers, BERT, LSTM/BiLSTM/GRU, Word2Vec, TF-IDF, Graph Neural Networks (GraphSAGE)
LLM Engineering: LLM fine-tuning (SFT, PEFT, LoRA, QLoRA), Unsloth, RAG (hybrid BM25 + dense retrieval, reranking), LangGraph, LangChain, Model Context Protocol (MCP), prompt-injection defense, Ollama
Backend and Web: FastAPI, Node.js, Express.js, React.js, REST APIs, Streamlit, MERN stack, asynchronous programming
Databases: MySQL, PostgreSQL (Supabase, pgvector), MongoDB, Neo4j, ChromaDB
Tools and DevOps: Git/GitHub, Docker, Linux, Kaggle, Google Colab

PROJECTS

AI Compliance Auditor (Llama-3 LoRA + RAG) 2026

- Built an end-to-end RAG system pairing a LoRA-fine-tuned Llama-3-8B with a hybrid BM25 and dense-embedding retriever reranked with FlashRank, restricted to policy-grounded compliance answers.
- Fine-tuned the model with Unsloth, PEFT and TRL SFTTrainer under 4-bit quantization on Kaggle T4 GPUs; built a Streamlit UI showing answers beside retrieved source chunks.
- Engineered prompt-injection defenses (pattern detection and keyword-overlap fallback) to reject manipulated or out-of-context queries.

Tech Stack: Unsloth, PEFT, TRL, BM25, FlashRank, Streamlit

Autonomous Multi-Agent AI Research Workspace 2026

- Built a multi-node research agent (planner, retriever, synthesizer, fact-checker) in LangGraph with human-in-the-loop review before web retrieval and automated verification of generated claims.
- Implemented a FastAPI backend with real-time token streaming and a Streamlit frontend, using AsyncSqliteSaver for stateful checkpointing and resumable workflows.

Tech Stack: LangGraph, FastAPI, Streamlit, SQLite

Omni-Support AI: Intelligent IT Support Agent 2026

- Developed a Model Context Protocol server exposing ticketing, knowledge-base search and web-search tools to a Groq-powered agent with multi-turn tool-calling, backed by Supabase (PostgreSQL + pgvector) and a role-based admin dashboard.
- Added screenshot analysis, Whisper voice input, hybrid vector and full-text knowledge-base search.

Tech Stack: Groq, MCP, Supabase (pgvector), Whisper

Enterprise Multi-Agent Knowledge Engine (Graph-RAG) 2026

- Designed a Dockerized Graph-RAG infrastructure combining Neo4j, ChromaDB and GPU-accelerated Ollama for local LLM inference.
- Established a QLoRA fine-tuning workflow on Kaggle with GGUF export and Ollama deployment via custom Modelfiles.

Tech Stack: Neo4j, ChromaDB, Ollama, Docker

GNN-Based BERT for Understanding Context from Music CSE425: Neural Networks

- Designed a multi-modal framework fusing a 3-layer GraphSAGE encoder over audio-segment graphs (140-dimensional Mel and Chroma node features) with a fine-tuned bert-base-uncased text encoder via cross-attention.
- Implemented multi-label genre tagging, valence/arousal regression and InfoNCE contrastive audio-text alignment; benchmarked against random, 2D-CNN and frozen-BERT baselines.

Tech Stack: PyTorch, GraphSAGE, BERT, librosa

Sarcasm Detection in News Headlines: ML vs Deep Learning Research Report

- Benchmarked Random Forest, SVM, BiLSTM and fine-tuned BERT on 26,709 headlines, achieving 84.87% accuracy and a 0.8248 F1-score.

Tech Stack: Random Forest, SVM, BiLSTM, BERT

Multi-Class Text Classification: Word Embeddings vs NN Models Report

- Compared TF-IDF with custom Skip-gram Word2Vec across nine models on 93,333 samples in 10 classes; GRU reached the best accuracy at 71.44%.

Tech Stack: TF-IDF, Word2Vec, RNN, LSTM, GRU

TravelBuddy: Travel Partner Matching and Itinerary Platform Full-stack

- Engineered an interactive travel itinerary platform with asynchronous API communication and state routing; designed MongoDB schemas isolating concurrent user accounts.

Tech Stack: MongoDB, Express.js, React.js, Node.js

WORK EXPERIENCE

Director, Department of Arts and Design September 2022 – Present
ROBU, Robotics Club of Brac University *Dhaka, Bangladesh*

- Lead a team of 75 people responsible for UI/UX direction, visual communication and structured documentation for robotics and automation initiatives and organized robotics events.
- Coordinate with hardware, software and management teams to plan, organize and manage exhibitions and national hackathons, supporting event delivery for large audiences.

RESEARCH

Modeling Pragmatic Implicature in LLMs Using Explicit Pragmatic Thought-Chains 2025 – 2026
Undergraduate Thesis, Brac University

- First author; developed a Pragmatic Inference Chain framework, grounded in Gricean Maxims, that helps language models decode sarcasm, indirect speech acts and scalar ambiguity.
- Applied supervised fine-tuning and PEFT (LoRA) on LLaMA and Mistral, evaluated with custom loss and interpretability metrics and validation confusion matrices.
- Supervised by Dr. Farig Yousuf Sadeque and Nirjhor Datta.

EDUCATION

BSc in Computer Science and Engineering September 2022 – October 2026
Brac University *Dhaka, Bangladesh*

- CGPA: 3.84 / 4.00 | Graduated with Highest Distinction

Higher Secondary Certificate (Pre-Engineering), GPA 5.00 / 5.00 2021
Govt. Azizul Haque College *Bogura, Bangladesh*

Secondary School Certificate, GPA 5.00 / 5.00 2019
Bogura Cantonment Public School and College *Bogura, Bangladesh*

HONORS, CERTIFICATIONS AND LANGUAGES

Vice Chancellor's Honor List (6 times) | Dean's Honor List (1 time) | Brac University
DataCamp (2026): Understanding Machine Learning; Cleaning Data in Python; Introduction to NumPy; Data Manipulation, Joining Data and Reshaping Data with pandas
Languages: Bangla (native) | English (professional proficiency)